

Steric Effects on the Chelation of Mn^{2+} and Zn^{2+} by Hexadentate Polyimidazole Ligands: Modeling Metal Binding by Calprotectin Site 2

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Abstract: Recently, there has been increasing interest in the design of ligands that bind Mn^{2+} with high affinity and selectivity, but this remains a difficult challenge. It has been proposed that the cavity size of the binding pocket is a critical factor in most synthetic and biological examples of selective Mn^{2+} binding. Here, we use a bioinspired approach adapted from the hexahistidine binding site of the manganese-sequestering protein calprotectin to systematically study the effect of cavity size on Mn^{2+} and Zn^{2+} binding. We have

designed a hexadentate, trisimidazole ligand whose cavity size can be tuned through peripheral modification of the steric bulk of the imidazole substituents. Conformational dynamics and redox potentials of the complexes are dependent on ligand steric bulk. Stability constants are consistent with the hypothesis that larger ligand cavities are relatively favorable for Mn^{2+} over Zn^{2+} , but this effect alone may not be sufficient to achieve Mn^{2+} selectivity.

Introduction

High-selectivity and high-affinity binding of metal ions is a critical feature for a range of biological functions and biomedical applications. For example, metalloproteins must be carefully tuned to select for the appropriate metal ion to fulfil their role, or complicated delivery and assembly systems must be leveraged to acquire the correct ion.^[1] For synthetic ligands, medicinal applications in chelation therapy and fluorescence tracking require both high affinity and high selectivity for the target ion, while applications involving imaging agents and some drugs require at least high affinity and low kinetic lability.^[2–6] More broadly, selective chelation of metal ions can be used to separate and recover valuable elements from their close cognates, for example, in the case of rare earth metals and actinides. However, some biologically relevant transition metal ions remain difficult to coordinate selectively, including Mn^{2+} . There are as yet only three (closely related) reported synthetic chelates that exhibit significant selectivity for Mn^{2+} over its close biological competitor, Zn^{2+} .^[7,8]

One area where nature has evolved potent strategies for strong and (in some cases) highly selective metal chelation is in metal sequestration as part of nutritional immunity, a strategy widely used by host immune systems as a defence against

pathogenic virulence by starving invading pathogens of necessary trace metal nutrients.^[9–11] In humans, the immune system protein calprotectin is a key player in this process, representing up to 60% of soluble cytosolic protein in neutrophils.^[12] This protein is understood to bind strongly to biologically relevant divalent first-row transition metals including Mn^{2+} , Zn^{2+} , Fe^{2+} , Cu^{2+} , and Ni^{2+} .^[13,14] Transition metal ion binding is strongly activated in the presence of high concentrations of Ca^{2+} ions, which coordinate to multiple sites in the protein and also act to induce heterotetramer formation.^[15–18] The complete picture of calprotectin's role in immunity and metal trafficking, and how it interacts with other proteins and systems involved in both host and pathogen metal homeostasis, is still developing. In particular, the understanding of calprotectin's role in metal sequestration across the first-row transition metals continues to evolve, and some of these interactions have been characterized only recently.^[13,16,19–23] However, the binding of calprotectin to Zn^{2+} and Mn^{2+} has been known for decades, and these ions are the most well-studied in this context, both in terms of their chemistry and structure, and their potential role in innate immunity.^[24–26] Because of the strong affinity of calprotectin for Mn^{2+} , and because of the unusual structure of its metal binding sites (see below), we have been interested in calprotectin as a possible model of how to tune a simple metal binding site for strong and potentially selective chelation of Mn^{2+} .

Calprotectin contains two transition-metal binding sites. Site 1 (Figure 1A) is an N_3O (His_3Asp) site that binds Zn^{2+} with strong affinity, but coordinates Mn^{2+} only relatively weakly ($\sim \mu\text{M}$ affinity).^[19,24] Site 2 (Figure 1B) strongly binds the whole range of biologically relevant first-row divalent transition metals at an N_6 (His_6) site, where two of the histidine donors are

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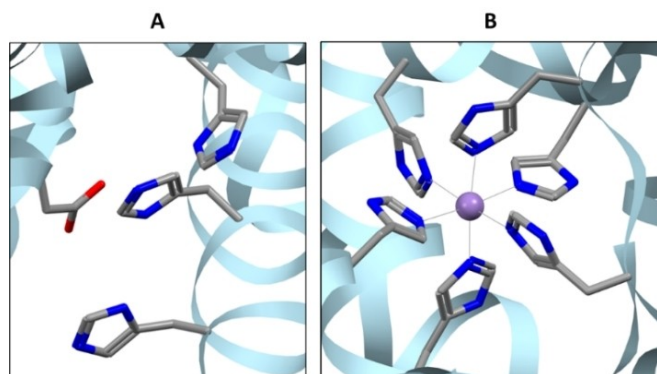


Figure 1. Crystal structure of the two metal coordination sites for calprotectin, A. apo-Site 1 (His, Asp, or N₃O) and B. Site 2 (His, or N₆) with manganese bound. (PDB code: 4XJK).^[17]

contributed by the C-terminal tail of the protein, which wraps around to encapsulate the transition metal upon binding.^[27]

To date, this is the only known naturally occurring His₆ metal binding site, and the only naturally occurring N₆ site known to coordinate Mn²⁺, with estimates of K_d on the order of ~1–100 nM.^[14,27] While precise estimates of the dissociation constants vary, the affinity of site 2 for divalent transition metals seems to follow the Irving–Williams series, which describes the order of thermodynamic stability of metal complexes in aqueous solution ($\text{Mn}^{2+} < \text{Fe}^{2+} < \text{Co}^{2+} < \text{Ni}^{2+} < \text{Cu}^{2+} > \text{Zn}^{2+}$).^[28,29] However, the affinities for Mn²⁺ and Zn²⁺ may be surprisingly close, with some estimates determined using the same method and the same conditions placing K_d for the two metals within an order of magnitude of each other; for example, measurements using isothermal titration calorimetry (ITC) gave $K_d = 1.3$ nM for Mn²⁺ and $K_d \geq 1.4$ nM for Zn²⁺, although other measurements have suggested that K_d for Zn²⁺ may actually be in the picomolar range.^[14,24] This contrasts with site 1, where all available data suggests that Zn²⁺ binds with a K_d at least 3 orders of magnitude lower than Mn²⁺, despite the N₃O site being more typical of common Mn²⁺ binding sites in metalloproteins.^[14,21,24] Because of its unconventional nature—contravening the common wisdom that Mn²⁺ binding sites generally contain a mix of oxygen and nitrogen donors—the Mn²⁺ binding site of calprotectin may suggest new approaches to the goal of designing chelators for Mn²⁺.^[21]

While calprotectin itself is not specifically selective for Mn²⁺, there are a few examples of metalloproteins with an apparent thermodynamic preference for Mn²⁺ over ions higher in the Irving–Williams series, including the Mn/Fe heterobimetallic R2-like ligand binding oxidase (R2lox) protein and the Mn/Mn class Ib ribonucleotide reductase R2 protein from *Bacillus anthracis*. In these cases, the affinities seem to depend on very subtle structural factors which have not been definitively identified.^[30–32] Recently, a selective protein-based sensor for Mn²⁺ was successfully developed by genetic modification of the lanthanide-binding protein lanmodulin, but structural characterization of the binding site was not yet reported, precluding definitive conclusions about the mechanism of

selectivity.^[2] These examples further suggest that model studies that could shed light on the origin of Mn²⁺ selectivity in biological and synthetic systems could be valuable.

Designing high-affinity and/or selective ligands for manganese(II) is challenging for a number of reasons. It lies at the bottom of the Irving–Williams series and has borderline hard/soft Lewis acid character, factors which make it difficult to target based on tuning the electronic properties (e.g. Lewis basicity) of the ligands. Because of its high-spin, d⁵ electron configuration, it has a spherical electron distribution and lacks any ligand field stabilization energy that would tend to favour a particular geometry. Therefore, neither ligand electronic properties nor shape-tuning knobs commonly used both in biological and synthetic systems to favour coordination of one transition metal over another—are particularly useful for selecting for Mn²⁺ over other biologically relevant divalent transition metals. However, this leaves open a third factor that could be used to help select for manganese: ligand cavity size and the resulting metal–ligand bond lengths. Of the biologically relevant divalent first-row transition metals, hexacoordinate high-spin Mn²⁺ has the largest ionic radius at approximately 97 pm; the ionic size decreases along the row, reaching 88 pm for hexacoordinate Zn²⁺.^[33] Therefore, a relatively large binding pocket—controlled and constrained by protein structure or by construction of a rigid ligand framework—could help favour Mn²⁺ binding.

In sufficiently constrained ligand systems, size effects can be a powerful method for tuning metal selectivity; for example, even small differences in ionic radius are routinely used to target selective chelates for lanthanide and actinide ions.^[34] This strategy has been relatively less explored for selective chelation of first-row transition metals, where efforts to tune binding affinities often focus on electronic factors and geometry rather than bond lengths. However, there have been a few recent successful examples of cavity-size-based selective binding of Mn²⁺. Cieslik et al. recently demonstrated this design principle with the use of a bispidine-based ligand (Figure 2A), which showed octadentate binding to Mn²⁺ while only binding Zn²⁺ in a hexadentate fashion, which was attributed to the smaller ionic radius of the Zn²⁺.^[7] The ligand was found to bind to Mn²⁺ more strongly than Zn²⁺ by 10–15 orders of magnitude, likely due to a combination of the higher denticity of Mn²⁺ binding

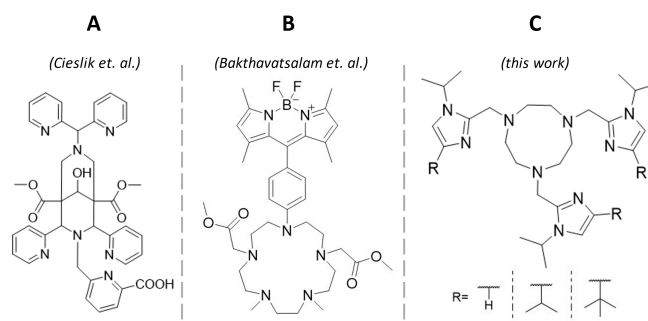


Figure 2. A. Octadentate ligand from Cieslik et al. showing high affinity for manganese(II).^[7] B. Heptadentate manganese(II)-activated fluorescent probe from Bakthavatsalam et al.^[3] C. Generalized ligand from this work (^{R1PR1L}).

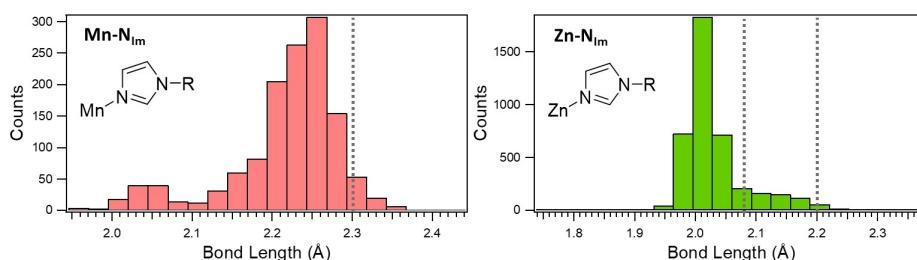


Figure 3. Histograms of Mn–N and Zn–N bond distances for coordinated imidazoles from the Cambridge Structural Database. The dashed vertical line in each case shows the (average) experimentally determined metal–N_{His} bond lengths for Mn²⁺ and Zn²⁺ in calprotectin's site 2.

and the overall large cavity size that better accommodated the larger size of Mn²⁺ over Zn²⁺. The larger size of Mn²⁺ over other divalent first-row transition metal ions has also been invoked as a design feature in a manganese-binding fluorescent probe with heptadentate coordination (Figure 2B).^[3] However, systematic studies of the effect of ligand cavity size on the selective binding of first-row transition metals in the absence of other effects (such as changes in donor types or denticity) are currently lacking.

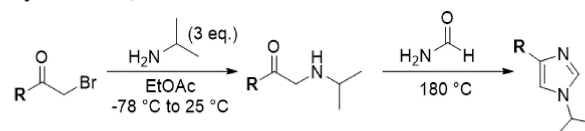
In the case of calprotectin, the presence of exclusively relatively soft histidine N-donor ligands may be expected to favour binding of the softer Lewis acid Zn²⁺, and the approximately octahedral geometry of the site should offer no particular advantage to Mn²⁺. However, in the highly constrained protein environment, it is possible that the binding site cavity size—i.e., imidazole-metal bond distances from the histidine donors—is tuned to help promote Mn²⁺ binding. Given this, we hypothesized that constraining imidazole-metal bond distances in a hexadentate calprotectin model ligand could be used to systematically vary the relative binding affinities of the ligand for different divalent first-row transition metals.

Support for this hypothesis can be found by examining the range of imidazole(N)-metal distances in molecular complexes of Mn and Zn tabulated in structures reported to the Cambridge Structural Database. The resulting histograms are shown in Figure 3. For manganese, the distribution of bond lengths is bimodal, with the smaller number of bond lengths below about 2.1 Å corresponding primarily to Mn³⁺ or higher oxidation state complexes. The majority of the Mn–N_{im} distances are distributed around approximately 2.25 Å, with a small but significant number of examples at or above 2.30 Å (67 Mn–N lengths $\geq$ 2.30 Å, or 4.6% of the total hitlist). On the other hand, the Zn–N_{im} bond lengths are strongly clustered between 1.95 and 2.05 Å (77.6% of hits occur in this range); few bond lengths are $\geq$ 2.20 Å (1.8% of hits), and only 0.3% of hits (13 examples) lie at $\geq$ 2.30 Å. Taken together, this suggests that if a chelating ligand were designed with sufficient steric pressure around the binding pocket that metal–N_{im} distances were constrained to be greater than about 2.20 Å, this could contribute significantly to favoring Mn²⁺ binding over Zn²⁺, in contravention of the Irving–Williams series. Experimental observations of metal–histidine bond lengths in calprotectin's site 2 suggest that this principle could be operative in this case; EXAFS data on calprotectin with Zn²⁺ bound in site 2 was best fit with four Zn–

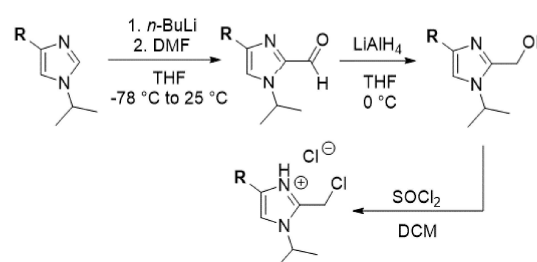
N_{His} distances of 2.20 Å and two Zn–N_{His} distances of 2.08 Å, suggesting that at least some of the Zn–N bonds are near the upper limit of favourable bonding interactions.^[27] On the other hand, the average Mn–N_{His} distances determined by X-ray crystallography are approximately 2.30(5) Å.^[26]

In order to test these hypotheses in a calprotectin-inspired synthetic scaffold, we envisioned a ligand platform that would exhibit a hexadentate N-donor coordination sphere, ideally including multiple imidazole donors as histidine mimics; that would be sufficiently constrained to allow deliberate tuning of the binding pocket; and that would permit for the ligand cavity size to be readily altered through peripheral substitution. The targeted ligand design is shown in Scheme 1, and is based on a 1,4,7-triazacyclononane (tacn) scaffold that imparts rigidity and pre-assembly of a three-fold symmetric coordination environment. Pendant imidazole donors complete the coordination sphere, and the substituent at the 4-position of the imidazole

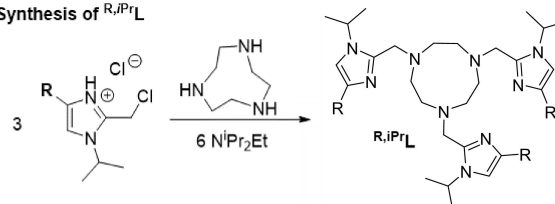
A. Synthesis of 1,4-disubstituted imidazoles



B. Synthesis of substituted 2-chloromethyl-imidazole hydrochlorides



C. Synthesis of R₃PtL



Scheme 1. Synthetic approach to substituted imidazoles and formation of the ligand R₃PtL.

rings is systematically varied from $R_4 = \text{H}$ to $R_4 = t\text{-butyl}$. As will be demonstrated below, increasing steric bulk at this position enforces a larger cavity size on the ligand due to repulsion between these groups, which are oriented towards the binding pocket. Manganese(II) and zinc(II) complexes of the ligands have been characterized structurally, spectroscopically, and electrochemically, providing insight into the effect of steric tuning on the properties of the compounds. Metal binding thermodynamics have been probed experimentally and computationally, supporting our overall hypothesis that a larger ligand cavity will decrease the gap between Mn^{2+} and Zn^{2+} binding affinities, although Zn^{2+} binding is still ultimately favoured in all cases.

Results and Discussion

Synthesis

In order to probe the impact of constrained binding pocket size on Mn^{2+} and Zn^{2+} chelation, we pursued a series of hexadentate ligands bearing three 4-substituted-N-isopropylimidazole donors arranged on a 1,4,7-triazacyclonane (tacn) anchor. While altering the substituent on the imidazole 1-nitrogen is relatively facile, substitution at this position unfortunately has little effect on the immediate environment around the metal center due to its peripheral location in the three-dimensional structure (see below). Therefore, for consistency, synthetic facility, and to provide a convenient spectroscopic handle, we used N-isopropyl substituted imidazoles in all cases. We focused instead on introducing different substituents at the 4-position, flanking the coordinating nitrogen of the imidazole rings and exerting a direct influence on the steric environment of the metal center. The introduction of increasingly bulky R groups in the 4 positions on the imidazole rings creates steric strain around the binding pocket, forcing the imidazole N-donors apart and increasing the minimum possible $\text{M}^{2+}\text{-N}_{\text{im}}$ bond distances. We note that two related ligands bearing N-methylimidazoles or unsubstituted N-H imidazoles, lacking 4-substituents ($R = \text{H}$), have previously been characterized and used to prepare coordination complexes of Fe, Ni, and Cu.^[35–38]

The synthetic pathway used to access the 4-substituted imidazoles and assemble the ligands is shown in Scheme 1. The 4-alkyl-N-isopropylimidazole precursors (Scheme 1A) were prepared using an adaptation of a previously reported route; all have been previously described.^[39] Preparation of the 2-chloromethylimidazole synthons (Scheme 1B) was accomplished by formylation of the corresponding imidazoles, followed by reduction using LiAlH_4 to the alcohol and chlorination with SOCl_2 ; these intermediates were isolated as their hydrochloride salts.^[40] The combination of these 2-chloromethylimidazole hydrochlorides with tacn in the presence of diisopropylethylamine (Scheme 1C) resulted in alkylation of each tacn nitrogen to give the final ligands. We note that an alternative synthetic route involving reductive amination of tacn with 2-imidazolecarbaldehyde precursors was also

successful (see Supporting Information). Metal complexation with each ligand was straightforwardly achieved on Mn^{2+} and Zn^{2+} by treatment of the ligand with one equivalent of the perchlorate salt of the metal cations in an ethanol solution. The resulting compounds were readily isolated and purified by crystallization from methanol/diethyl ether or acetonitrile/diethyl ether. Detailed procedures and characterization data are given in the Supporting Information.

Structural Characterization

Mn^{2+} and Zn^{2+} complexes of H_4iPrL , iPr_2iPrL , and tBu_2iPrL have been structurally characterized as their perchlorate salts. $[\text{tBu}_2\text{iPrL}\text{Mn}](\text{ClO}_4)_2$ is shown in Figure 4, while the structures of $[\text{H}_4\text{iPrL}\text{Mn}](\text{ClO}_4)_2$ and $[\text{iPr}_2\text{iPrL}\text{Mn}](\text{ClO}_4)_2$ along with the corresponding Zn^{2+} complexes can be found in the Supporting Information (Figure S51–S53). All structures reveal the expected hexa-coordinate, pseudooctahedral metal centers, with 3-fold or approximate 3-fold symmetry. For both metal ions, complexes of H_4iPrL and tBu_2iPrL exhibit crystallographic 3-fold symmetry with one-third of the complex occupying the asymmetric unit; for these ligands, complexes of the two metals are isomorphic. In contrast, the structures of the iPr_2iPrL complexes both show two full independent complexes in the asymmetric unit, and there is significant asymmetry in the metal-ligand bond distances, especially for $[\text{iPr}_2\text{iPrL}\text{Zn}]^{2+}$; in this case, the Mn^{2+} and Zn^{2+} complexes crystallize in different space groups (See Table S3). Space-filling models shown in Figure 5 for all three manganese(II) complexes illustrate the steric congestion around the metal center that results when bulkier R_4 groups are

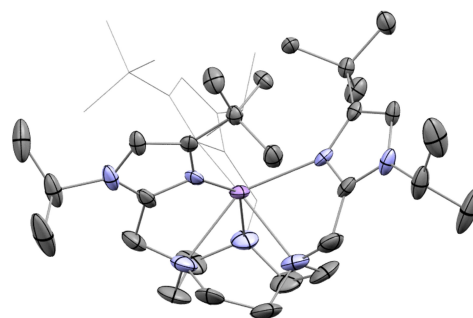


Figure 4. Crystal structure of $[\text{tBu}_2\text{iPrLMn}](\text{ClO}_4)_2$.^[41] Thermal ellipsoids are shown at 50%; perchlorate counteranions and hydrogen atoms are omitted for clarity. (Color code: blue = N, gray = C, magenta = Mn).

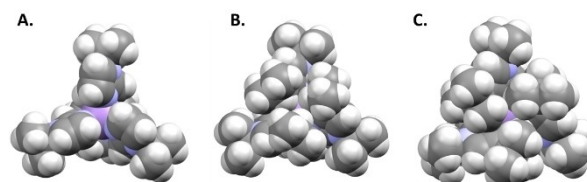


Figure 5. Space filling models for A) $[\text{H}_4\text{iPrLMn}]^{2+}$, B) $[\text{iPr}_2\text{iPrLMn}]^{2+}$, and C) $[\text{tBu}_2\text{iPrLMn}]^{2+}$. Perchlorate counter anions and cocrystallized acetonitrile solvent are omitted for clarity.

introduced to the imidazoles. The crystallographically determined M^{2+} -N_{im} and M^{2+} -N_{tacn} bond distances are summarized in Table S2 and shown in Figure 6.

As expected, the metal-imidazole bond distances systematically increase as the steric bulk of the 4-substituent increases, $R=H < {}^i\text{Pr} < {}^t\text{Bu}$. Metal-N_{im} bond distances are shorter for Zn^{2+} than for Mn^{2+} when $R=H$ or ${}^i\text{Pr}$, but this difference nearly disappears for $R={}^t\text{Bu}$, suggesting that for this more sterically strained cavity, ligand bulk dominates in determining the structure. This may be reminiscent of the situation in a protein such as calprotectin where the size and shape of the metal binding cavity are likely to be highly constrained by the protein structure. Interestingly, the Zn-N_{tacn} distances for the ${}^t\text{Bu}_3\text{iprL}$ complex are also anomalously short, perhaps as a compensatory effect due to the weaker binding of the imidazole nitrogens in this complex; other than this anomaly, there is significantly less variation in the M-N_{tacn} distances than in the M-N_{im} distances (Figure 6). At 2.296(2) Å, the Zn-N_{im} bond length in $[\text{tBu}_3\text{iprLZn}]^{2+}$ is one of the longest crystallographically

determined zinc-imidazole distances to date in a small molecule (see Figure 3).

In the case of $[\text{ipr}_3\text{iprLZn}]^{2+}$, the plots in Figure 6 also highlight the significant asymmetry of the bond distances around Zn^{2+} , possibly as a result of steric constraints preventing the ideal Zn-N_{im} bond lengths from being satisfied in all cases. The corresponding Mn-N_{im} distances in $[\text{ipr}_3\text{iprLMn}]^{2+}$ exhibit a much smaller variation.

NMR Characterization and Conformational Dynamics in Solution

While the high-spin (μ_{eff} (Evans Method) = 5.7, 6.2, and 5.9 μ_B for $R=H$, ${}^i\text{Pr}$, and ${}^t\text{Bu}$, respectively) paramagnetic $[\text{R}_3\text{iprLMn}]^{2+}$ complexes showed only a few detectable extremely broad resonances by ${}^1\text{H}$ NMR, $[\text{R}_3\text{iprLZn}]^{2+}$ complexes are diamagnetic and could be characterized by ${}^1\text{H}$ and ${}^{13}\text{C}\{^1\text{H}\}$ NMR spectroscopy. The ${}^1\text{H}$ NMR spectra reveal dynamic processes that give insight into ligand conformational dynamics and their dependence on the steric bulk of the imidazole 4-substituents. All complexes show three-fold symmetry in solution, where only one resonance or set of resonances corresponding to the imidazole rings are observed. The 12 methylene protons of the tacn ring are chemically and magnetically equivalent in the free ligand, manifesting as one singlet in the ${}^1\text{H}$ NMR; upon coordination to a metal ion, this equivalency is broken. In the spectrum of the less hindered $[\text{H}_3\text{iprLZn}]^{2+}$ complex, the tacn methylene protons split into two multiplets near 2.8 and 3.0 ppm (Figure 7A), one corresponding to H_a and H_b and the other corresponding to H_c and H_d (Figure 7B). The conformational dynamics of the ligand are rapid enough on the NMR timescale to exchange the axial and equatorial positions on the ring. By contrast, in the

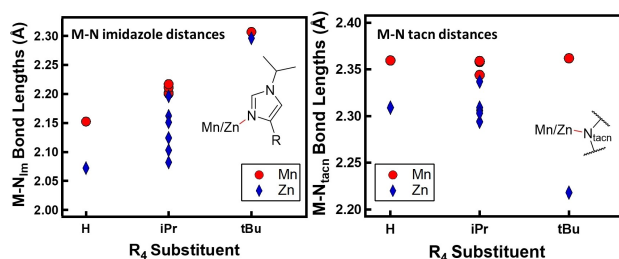


Figure 6. M^{2+} -N distances for the imidazole nitrogens and for the tacn nitrogens are plotted. Solid red circles ($\bullet$, Mn^{2+}) and blue diamonds ($\blacklozenge$, Zn^{2+}) correspond to crystallographically determined bond distances.

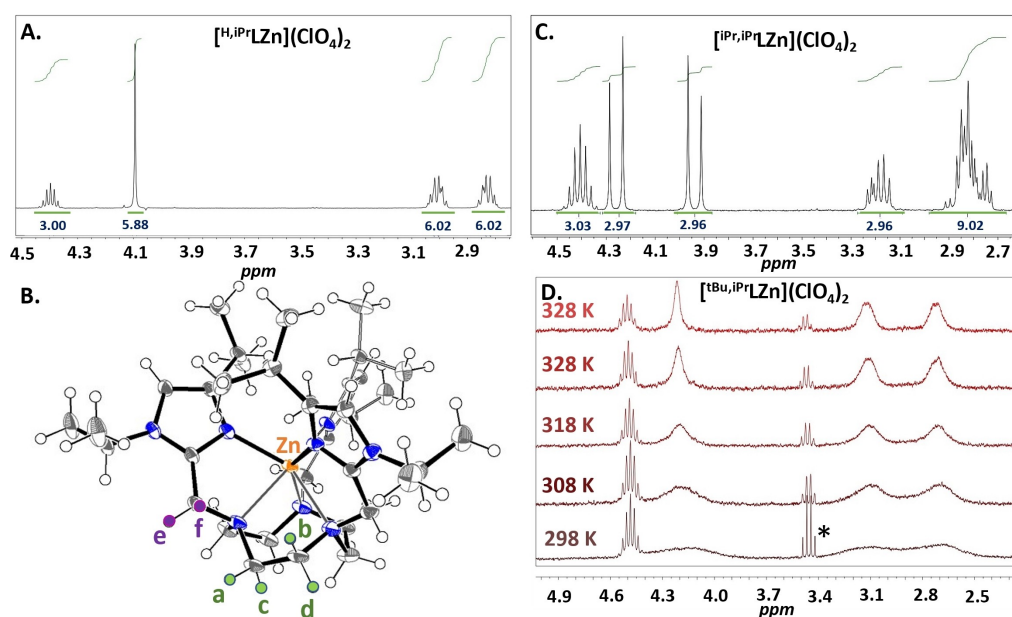


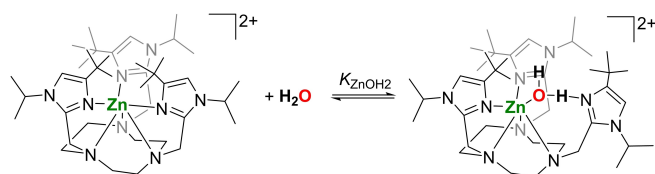
Figure 7. A. ${}^1\text{H}$ NMR of $[\text{H}_3\text{iprLZn}](\text{ClO}_4)_2$ in CD_3CN at room temperature. B. Hydrogen labeling of ligand, with $[\text{ipr}_3\text{iprLZn}]^{2+}$ shown as example. C. ${}^1\text{H}$ NMR of $[\text{ipr}_3\text{iprLZn}](\text{ClO}_4)_2$ in CD_3CN at room temperature. D. Variable temperature ${}^1\text{H}$ NMR of $[\text{tBu}_3\text{iprLZn}](\text{ClO}_4)_2$ in CD_3CN (*EtOAc).

spectrum of the more sterically constrained $[\text{ipr}_2\text{PrLZn}]^{2+}$ complex (Figure 7C), these dynamics are apparently slowed significantly; now, four inequivalent (overlapping) multiplets are apparent for H_a , H_b , H_c , and H_d , suggesting that exchange between the axial and equatorial positions is slow on the NMR timescale. Furthermore, the protons of the imidazolyl- CH_2 -tacn bridge (H_e and H_f in Figure 7B) have now become magnetically inequivalent due to their different positions in the “frozen” ligand conformation, as have the methyl groups on the isopropyl substituents. Variable temperature NMR (Figure S24) up to 65°C results only in slight broadening of the peaks but is insufficient to reach coalescence, suggesting that the barrier to conformational exchange is relatively high.

The ^1H NMR spectrum of the $[\text{tBu}_2\text{iprLZn}]^{2+}$ complex also shows evidence for slow dynamics, but in this case the coalescence temperature for most resonances appears to be below room temperature and the resonances show significant broadening at 25°C ; we estimate the rate constant at this temperature to be approximately 230 s^{-1} . Heating to 65°C partially sharpens the resonances but does not reach the fast-exchange limit (Figure 7D). This suggests that there is a lower barrier for conformational exchange in this ligand than in ipr_2PrL , despite the increased steric bulk of the tBu substituents; we hypothesize that this is the result of the significantly longer Zn–N distances in $[\text{tBu}_2\text{iprLZn}]^{2+}$ as observed in the crystal structure. This may create greater flexibility around the metal center; alternatively, weaker binding of the imidazole donors may facilitate a dissociative mechanism of conformational exchange involving transient decoordination of one of the imidazole arms.

Ligand Hemilability and Binding of Water to $[\text{tBu}_2\text{iprLZn}]^{2+}$

Addition of water to CD_3CN solutions of $[\text{tBu}_2\text{iprLZn}](\text{ClO}_4)_2$ resulted in the appearance of a new set of ^1H NMR peaks which are also broadened by dynamic processes at room temperature (Figure S28). In addition to new signals attributable to the ligand, a peak appears at 9.7 ppm. 2D EXSYNMR experiments (Figure S30) suggest that this peak is in chemical exchange with water because of the positively phased correlation peak between the water signal at 2.17 ppm and the new peak at 9.7 ppm. We attribute this to the formation of a new species where water has coordinated to the Zn^{2+} center, displacing one of the imidazole donor arms of the ligand. The coordinated water molecule would likely engage in hydrogen bonding to the uncoordinated imidazole (Scheme 2), which is supported by



Scheme 2. Equilibrium water binding and imidazole hemilability on $[\text{tBu}_2\text{iprLZn}]^{2+}$, showing proposed structure of water-bound complex.

DFT calculations (Figure S39); however, we cannot completely rule out experimentally that the complex may be better described as a protonated pendant imidazole hydrogen bonded to a coordinated hydroxide ion. The hydrogen bonding may promote this ligand exchange by stabilizing the water complex. Given the preorganized nature of the ligand binding site, and the fact that the chelate effect should disfavor partial ligand decoordination, the ability of a water molecule to displace one of the ligand arms is somewhat unexpected. Since this behavior is not observed for the less hindered $[\text{ipr}_2\text{PrLZn}]^{2+}$ system (Figure S29), this is presumably due to the overly large ligand cavity which precludes ideal bond lengths around Zn^{2+} , resulting in the hemilability of the imidazole donors. Ligand hemilability has been shown to be a useful property of metal catalysts for a range of reactions.^[42]

The water binding is an equilibrium process (Scheme 2) with K_{ZnOH_2} estimated to be 85 ± 12 at room temperature based on NMR titration studies (Figure S28). The process appears to be slow on the NMR time scale at room temperature.

Electrochemical Characterization: Steric Tuning of Redox Potentials

The redox properties of the compounds were probed by cyclic voltammetry in acetonitrile. The zinc complexes showed no evidence of accessible redox processes in the range from -2.0 to $+1.0\text{ V}$ vs. Fc/Fc^+ (Figures S32–S37). In contrast, $[\text{H}_2\text{iprLMn}]^{2+}$ showed two clear oxidative features (Figure 8). The first feature is quasi-reversible, with $E_{\text{p,a}} = 0.83\text{ V}$ and $E_{\text{p,c}} = 0.26\text{ V}$ for the anodic and cathodic peaks, respectively; this is attributed to an $\text{Mn(II)}/\text{Mn(III)}$ couple and the quasi-reversibility suggests that electron transfer to/from the electrode is slow. Slow electron transfer in $\text{Mn(II)}/\text{Mn(III)}$ couples is commonly observed and generally attributed to significant geometric changes upon oxidation due to Jahn–Teller distortion in Mn(III) .^[43] Scanning further oxidatively, a reversible redox peak observed at $E_{1/2} = 1.23\text{ V}$ vs. Fc/Fc^+ is attributed to an $\text{Mn(III)}/\text{Mn(IV)}$ couple.

The more sterically hindered $[\text{ipr}_2\text{PrLMn}]^{2+}$ shows a similar quasi-reversible $\text{Mn(II)}/\text{Mn(III)}$ couple with the redox features shifted anodically to $E_{\text{p,a}} = 1.04\text{ V}$ and $E_{\text{p,c}} = 0.52\text{ V}$. No $\text{Mn(III)}/$

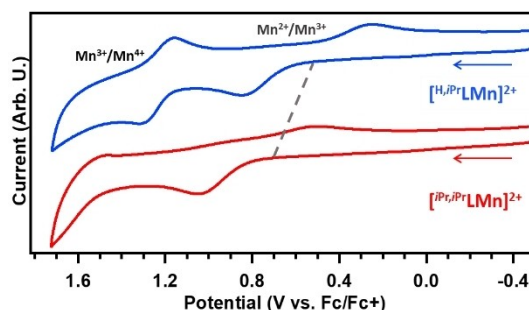


Figure 8. Cyclic voltammograms for manganese compounds with $\text{R}=\text{H}$, ipr , tBu . Data was collected in 0.1 M tetrabutylammonium hexafluorophosphate in degassed acetonitrile under an argon atmosphere. The concentration of the compounds was 1 mM .

Mn(IV) couple is clearly observed within the measurement window, but could be shifted to higher potentials outside this range. In the case of $[\text{tBu}_2\text{iPrLMn}]^{2+}$, no clear oxidation features are observed (Figure S34), even at slow scan rates (25 mV/s); in this case, we cannot definitively say if the lack of an Mn(II)/Mn(III) feature is because its potential has been shifted outside the measurement window, or due to very slow electron transfer kinetics.

The shift to higher oxidation potentials for $R_4 = \text{iPr}$ as compared to $R_4 = \text{H}$ is unexpected at first glance; one might expect a small shift in the opposite direction (i.e., making $[\text{iPr}_2\text{iPrLMn}]^{2+}$ easier to oxidize) due to the more electron-donating alkyl substituents. We hypothesize that this is due to the larger binding pocket of iPr_2iPrL , which does not as readily accommodate the shorter Mn(III)-N bond lengths, making this an example of *steric*, rather than electronic, tuning of redox properties.^[44,45]

Stability Constant Determinations

Potentiometric titrations were used to determine the protonation constants of the ligands and the stability constants of the metal complexes; results are shown in Table 1 for H_2iPrL , iPr_2iPrL , and tBu_2iPrL . Protonation constants shown in the table correspond to the general expression shown in Equation (1). Titrations were carried out at 25 °C under a blanket of inert gas in a 60:40% v/v mixture of water and acetonitrile (with 0.15 M NaCl to maintain approximately constant ionic strength) due to the low solubility of the ligands and some of the complexes in pure water; the data are therefore not directly comparable to binding constants measured in water.



For each ligand, four protonation constants were measured in the absence of metal ions over the pH range from 2 to 12 (Figures S40–S42). Although the ligands potentially have up to

	$R_4 = \text{H}$	$R_4 = \text{iPr}$	$R_4 = \text{tBu}$
$\log(K_{\text{H1}})$	8.99(5) ^[a]	8.86(9)	8.73(8)
$\log(K_{\text{H2}})$	7.07(9)	7.11(4)	7.15(16)
$\log(K_{\text{H3}})$	5.33(5)	5.69(4)	5.28(6)
$\log(K_{\text{H4}})$	1.8(4)	2.1(3)	2.4(3)
$\log(K_{\text{ZnH}})$	—	—	5.3(3)
$\log(K_{\text{Zn}})$	> 24	> 24	10.4(1)
$\log(K_{\text{Mn}})$	14.7(8)	14.9(2)	7.1(5)
$\log(K_{\text{Zn}}/K_{\text{Mn}})$	> 9.3	> 9.1	3.3(5)
Calc. $\log(K_{\text{Zn}}/K_{\text{Mn}})$ ^[b]	10.7	7.3	3.7
pZn ^[c]	> 12.8	> 12.4	5.5
pMn ^[c]	7.9	7.9	4.0

[a] Numbers given in parentheses correspond to standard deviation on the last digit shown. Standard deviations are given either based on the average of at least three independent trials, or on the standard deviation of the least-square refinements used to fit the data, whichever was larger.
[b] B3LYP-GD3, 6-311g(d). [c] pM calculated for $C_{\text{M}} = C_{\text{L}} = 2 \times 10^{-3}$ M, pH = 7.4.

six basic sites, attempting to fit titration data to more than four protonation constants did not improve the fit, suggesting that in this pH range protonation states beyond H_4L^{4+} are not accessed. All four protonation constants are very similar for the three ligands studied. By analogy to other tacn-based ligands, and based on the greater basicity of tertiary alkylamines as compared to imidazoles, the first protonation at approximately $\log(K_{\text{H1}}) \sim 8.9$ is believed to occur on one of the tacn tertiary alkylamine sites.^[46]

After the first protonation, subsequent protonation of the tacn ring amines becomes more difficult due to electrostatic repulsion.^[47] The second protonation (K_{H2}) is attributed to protonation on one of the imidazole rings, given that the pK_{a} of the conjugate acid of imidazole is approximately 7.

Speciation diagrams for ligands H_2iPrL and tBu_2iPrL in the presence of one equivalent of Zn^{2+} under the conditions described above are shown in Figure 9 (corresponding diagrams for iPr_2iPrL are given in Figure S41). The stability constants determined for $[\text{R}_2\text{iPrLZn}]^{2+}$ complex formation are given in Table 1. The formation constants for $[\text{H}_2\text{iPrLZn}]^{2+}$ and $[\text{iPr}_2\text{iPrLZn}]^{2+}$ are high, corresponding to essentially complete formation of the complex at $\text{pH} \geq 2$; because of this, the binding constant could not be determined exactly by these measurements and only a lower limit is given ($\log(K_{\text{Zn}}) > 24$). Previously reported tacn-anchored chelating ligands bearing amine donors in place of imidazoles have shown similarly high binding constants (e.g., $\log(K_{\text{Zn}}) = 21.3$).^[48] The stability of the $[\text{tBu}_2\text{iPrLZn}]^{2+}$ complex is

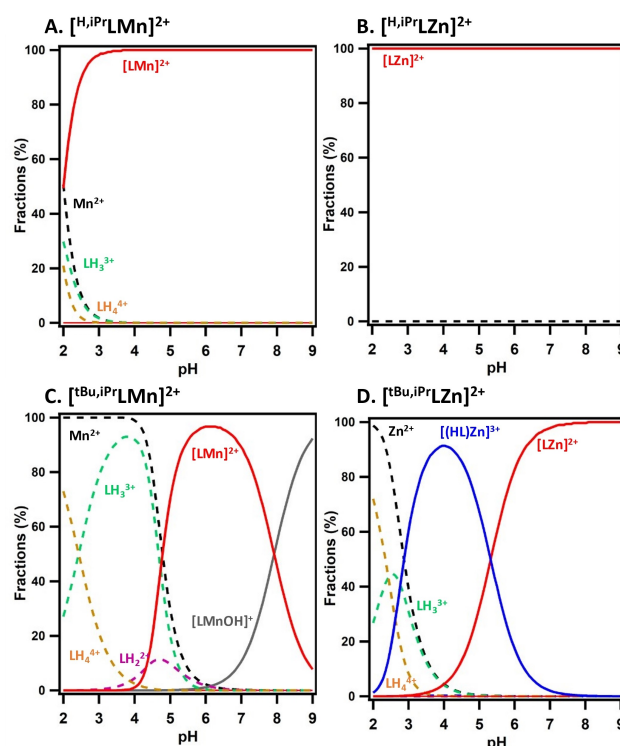
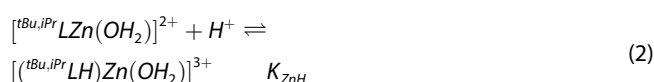


Figure 9. Species distribution curves for binding of H_2iPrL with Mn^{2+} (A) and Zn^{2+} (B, calculated using the lower limit value of K_{Zn}) and for binding of tBu_2iPrL with Mn^{2+} (C) and Zn^{2+} (D). Note: coordinated waters are omitted from formulae in species labels. Distribution curves were simulated for $[\text{L}] = [\text{Mn}^{2+}] = [\text{Zn}^{2+}] = 3$ mM.

significantly lower, with $\log(K_{\text{Zn}}) = 10.4$. This is reflected in the speciation diagram, with significant amounts of uncoordinated ligand observed at low pH (<5). At a pH of 12, the slow formation of zinc hydroxide and/or oxide precipitates was observed. In fitting the titration data for the $[\text{tBu}_2\text{iPrLZn}]^{2+}$ system, an additional equilibrium constant was included (K_{ZnH} , Equation (2)) for the protonation of the ligand at a pendant, uncoordinated imidazole on the basis of the NMR data discussed above, which suggested that an imidazole donor arm can be displaced by water. The resulting speciation diagram shows that this protonated $[(\text{tBu}_2\text{iPrLH})\text{Zn}(\text{OH}_2)]^{3+}$ species dominates in solution between roughly pH=3 to pH=5. No evidence for this behavior was observed for any of the other Zn^{2+} complexes. The inclusion of an equilibrium process for the formation of a hydroxide complex (i.e., $[\text{tBu}_2\text{iPrLZnOH}]^+$) was tested, but did not improve the fit.



Complexation of Mn^{2+} with R_4iPrL was slow, requiring a batch titration approach to study the equilibrium (Figures S46—S48). Despite the lability of Mn^{2+} in general, slow complex formation with chelating ligands has been observed in a number of cases, and has in at least one case been shown to result from the pre-equilibrium formation of an unstable “pre-complex” that more slowly converts to the stable complexed product.^[7,49] This may also be the case in our system, but we have not attempted to probe this process further. By analogy to the $[\text{tBu}_2\text{iPrLZn}]^{2+}$ system, we considered the possible presence of a ligand-protonated water complex, $[(\text{tBu}_2\text{iPrLH})\text{Mn}(\text{OH}_2)]^{3+}$; however, this did not improve the fit and preliminary relaxivity measurements are consistent with a lack of water binding (see Supporting Information, Figure S31). However, including an equilibrium for the formation of a manganese hydroxide complex, for example $[\text{tBu}_2\text{iPrLMnOH}]^+$, did significantly improve the fit for this system (see Supporting Information); simulations suggest that such a species may be present in significant amounts in solution at pH > 6.

For all ligands, the Mn^{2+} complex is less stable than the Zn^{2+} complex, in line with the Irving–Williams series. The strongest binding is observed for H_4iPrL , with $\log(K_{\text{Mn}}) = 14.7$. While this value is significantly lower than that determined (under slightly different conditions) for the most stable known Mn^{2+} complexes (e.g., an 8-coordinate bispidine ligand with $\log(K_{\text{Mn}}) \sim 25$ in water)^[7] it compares favorably with many well-known chelating ligands such as ethylenediaminetetraacetic acid (edta) and 1,2-bis(*o*-aminophenoxy)ethane-*N,N,N',N'*-tetraacetic acid (bapta), with $\log(K_{\text{Mn}}) = 12.46$ and 9.14, respectively, although these values should be compared with caution as they were determined in different solvents.^[50,51] As with Zn^{2+} , binding to tBu_2iPrL is significantly weaker than the less hindered ligands for Mn^{2+} ($\log(K_{\text{Mn}}) = 7.1$).

The difference between the Zn^{2+} and Mn^{2+} binding constants is tabulated as $\log(K_{\text{Zn}}/K_{\text{Mn}})$, and decreases significantly as the size of the R_4 substituent increases from $\text{R}_4 = \text{H}$ to

$\text{R}_4 = \text{t-Bu}$. In particular, while the less hindered ligands H_4iPrL and iPr_2iPrL bind Zn^{2+} more strongly than Mn^{2+} by at least 9 orders of magnitude in K_{Mn} , the tBu_2iPrL ligand favors zinc by only 3.3 orders of magnitude, more similar to the situation in calprotectin site 2. This is consistent with our hypothesis that increasing the size of the binding pocket by tuning the steric bulk at R_4 would tend to favor Mn^{2+} coordination over Zn^{2+} ; however, within this series this change was insufficient to completely overturn the selectivity. Furthermore, this improvement comes at the expense of a much lower association constant for each metal and hemilability of one of the donor arms.

Computational Studies

Cavity size calculations were carried out using empirical force field calculations, using an adaptation of an established method, in order to further illustrate the influence of the R_4 substituents on tuning metal ion binding through cavity size manipulation (see Supporting Information for further details).^[7,52,53] In this method, the metal-ligand bond distances are constrained to specific values and systematically varied, while the ligand steric energy is calculated as a function of these M–L distances. The energies of the metal-ligand bonds are not themselves included in the calculation, making the results independent of the metal ion identity. Here, we considered different M– N_{tacn} and M– N_{im} distances, but all three M– N_{tacn} distances were constrained to the same value, and all three M– N_{im} distances were constrained to the same value at each point in the calculation (i.e., approximate three-fold symmetry was maintained). This is likely to be a reasonable simplification since in most cases (with the significant exception of $[\text{iPr}_2\text{iPrLZn}]^{2+}$) the crystallographically determined structure is three-fold symmetric or approximately so. Plotting the ligand steric energy as a function of the constrained distances gives rise to the surfaces shown in Figure 10 for $\text{R}_4 = \text{H}$, iPr , and t-Bu . The coordinates of the crystallographically determined bond lengths for Zn^{2+} and Mn^{2+} are shown on the surface as green and red dots, respectively.

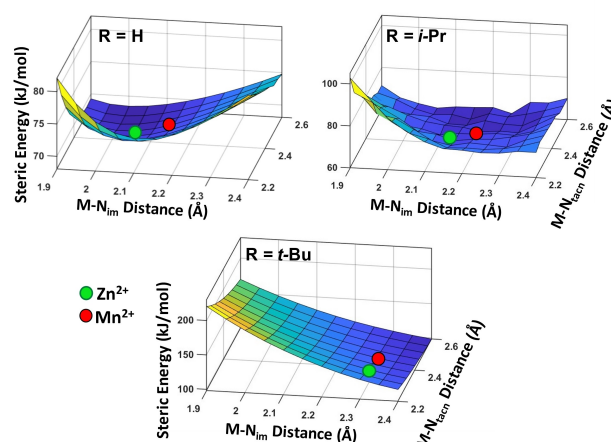


Figure 10. Steric energy vs. bond lengths for H_4iPrL , iPr_2iPrL , and tBu_2iPrL .

For $R_4 = \text{H}$, the ligand cavity size is the smallest, and the surface shows a relatively shallow bowl with a minimum near $M\text{-N}_{\text{im}} = 2.0 \text{ \AA}$ and $M\text{-N}_{\text{tacn}} = 2.5 \text{ \AA}$; the experimental bond lengths for the Zn^{2+} complex lie at a slightly lower energy than those for Mn^{2+} , but the difference in this case is small. For $R_4 = \text{'Pr}$, there is a clear tilt of the surface towards longer bond lengths; the corrugated nature of the surface may be due to the shape of the isopropyl groups, which tends to result in "gearing" and more complicated intraligand interactions. In this case, both Mn^{2+} and Zn^{2+} structures lie near the minimum. Finally, for $R_4 = \text{'Bu}$, steric strain has a much larger effect, especially at short bond distances, and the surface slopes towards the longest tested bond lengths ($M\text{-N}_{\text{im}} = 2.4 \text{ \AA}$ and $M\text{-N}_{\text{tacn}} = 2.6 \text{ \AA}$) without reaching a minimum. The experimentally determined Mn^{2+} bond lengths are found slightly downhill of those for Zn^{2+} by approximately 11 kJ/mol (2.6 kcal/mol), which mostly results from the longer $M\text{-N}_{\text{tacn}}$ distances in the Mn^{2+} complexes, since the $M\text{-N}_{\text{im}}$ distances are more similar between the two.

We further probed the thermodynamics of ligand binding using DFT calculations using Gaussian 16.^[54] Geometry optimizations were first carried out starting from the crystallographically determined coordinates. Unfortunately, a wide range of tested DFT methodologies struggled to faithfully reproduce the geometry of some of the complexes, especially $[\text{'Bu}_4\text{PrLZn}]^{2+}$, underestimating the Zn-N_{im} bond lengths and overestimating the $\text{Zn-N}_{\text{tacn}}$ bond lengths, often by more than 0.15 Å. Optimization using B3LYP-D3/6-311 g(d) gave reasonable agreement with the experimental bond lengths for most complexes (Figure S38) and thermodynamic calculations gave computed $\log(K_{\text{Zn}}/K_{\text{Mn}})$ values in good agreement with experiment (Table 1).^[55] The computational results overall support our hypotheses about the impact of steric bulk and cavity size on Zn^{2+} and Mn^{2+} binding.

Conclusion

Tuning the selectivity of chelating ligands for Mn^{2+} over other divalent first-row transition metals is a technologically important problem, and the factors involved in achieving high Mn^{2+} selectivity in many systems have not been fully elucidated. The work reported here addresses one aspect of this question using a biomimetic approach, inspired by the protein calprotectin. A series of hexadentate tacn-based ligands with systematically increasing cavity size due to remote steric pressure from bulky pendant imidazoles was synthesised in order to systematically probe cavity size effects on Mn^{2+} and Zn^{2+} binding and on the properties of the resulting complexes. Experimental and computational studies show that increasing the cavity size had a more significant detrimental effect on binding of the smaller Zn^{2+} ion relative to Mn^{2+} , although complete disruption of the Irving–Williams series to favour Mn^{2+} binding was not achieved on the current scaffold. Additionally, steric effects on the Mn^{2+} redox potentials and on the ligand conformational dynamics in the Zn^{2+} complexes are demonstrated. While this work validates the importance of cavity size on achieving selective

metal binding among first-row transition metals, it also suggests that this factor alone may be insufficient to select for Mn^{2+} ; instead, strategies involving higher-denticity donors (e.g. heptadentate and octadentate ligands) are likely necessary, and are currently the target of further investigation by our research group and others.

Experimental Section

Materials and instruments: N-isopropylimidazole was purchased from Accela, 1,4,7-triazacyclononane (tacn) and diisopropylethylamine (DIPEA) was purchased from and Tokyo Chemical Industry, and hexahydrate perchlorate metal salts at 99% purity for manganese(II) and zinc(II) were purchased from Acros Organics and Alfa Aesar respectively. 1-isopropyl-4-tertbutylimidazole and 1,4-diisopropylimidazole were synthesized as previously reported.^[39,56–58] Any other starting materials, reagents, and solvents were purchased from standard chemical vendors and used without further purification, except where described otherwise below.

Electrospray ionization high-resolution mass spectrometry (ESI-HR-MS) was performed on a Bruker micrOTOF-QII via injection of samples as acetonitrile solutions. Electron impact mass spectrometry (EI-MS) was performed via a Shimadzu GCMS-QP2010 Ultra using samples introduced as dichloromethane solution. Cyclic voltammetry (CV) was performed using a CHI 600E potentiostat using a Pt wire counter-electrode and an Ag/Ag^+ pseudoreference electrode. The glassy carbon working electrode was polished with 0.05 μm alumina powder. Acetonitrile was used as the solvent, containing 0.1 M Bu_4NPF_6 electrolyte which had been purified by recrystallization from ethanol. All samples were measured at 0.1 mM concentrations. Elemental analysis was performed using an Elementar UNICUBE analyzer which was standardized using sulfanilamide, or by Midwest Microlabs (Indianapolis, IN). ^1H NMR and $^{13}\text{C}\{^1\text{H}\}$ NMR were measured on Bruker AVANCE III NMRs equipped with 300, 500, or 600 MHz superconducting magnets. NMR data were referenced internally to residual solvent, and all NMR spectra were collected at room temperature (298 K) except where otherwise stated.

General procedure for the synthesis of 4-alkyl-1-isopropylimidazole compounds: The disubstituted imidazoles were synthesized according to a previously reported procedure^[39,56] starting from α -bromoketone precursors.^[58] The imidazoles were purified by vacuum distillation, and the spectroscopic properties were consistent with previous reports.

General procedure for the synthesis of 4-alkyl-1-isopropylimidazole-2-carbaldehyde compounds: The synthesis was adapted from previously reported routes used to prepare related compounds.^[59,60] The 4-alkyl-1-isopropylimidazole precursor (1 equiv.) was dissolved in a Schlenk flask in dry THF at a concentration of approximately 0.15 M and placed under inert gas on a Schlenk line. This flask was cooled to -78°C . *n*-Butyllithium (1.6 M in hexanes, 1.5 equiv.) was added dropwise, and the flask was allowed to stir at -78°C for 1.5 h. Anhydrous DMF (1.75 equiv.) was then added to the stirred solution, which was stirred for 4 h at -78°C , and allowed to slowly warm to room temperature overnight. Deionized (DI) water (approximately half the reaction volume) was added to the reaction flask, and the mixture was transferred to a separatory funnel, where it was extracted with diethyl ether three times. The organic layers were combined and washed with water, and then brine, before drying with anhydrous sodium sulfate. The sodium sulfate was removed by filtration, and the organic layer evaporated, revealing a yellow oil of crude product. In most cases, the crude product could be used in the next step without further purification.

1-isopropylimidazole-2-carbaldehyde: The oil was crystallized from a concentrated solution in diethyl ether, cooled in a freezer overnight, affording clear colorless crystals (27.27 g, 46%). ^1H NMR (500 MHz, CDCl_3 , 25 °C): δ = 9.84 (s, 1H), 7.34 (s, 1H), 7.32 (s, 1H), 5.49 (sept, J = 10 Hz, 1H), 1.49 (d, J = 10 Hz, 6H) ppm. Spectroscopic data was consistent with previous reports.^[40,61]

1,4-diisopropylimidazole-2-carbaldehyde: The crude product was an oil which was used for the next step without further purification (3.14 g, 94%). ^1H NMR (300 MHz, CDCl_3 , 25 °C): δ = 9.77 (d, J = 0.7 Hz, 1H), 7.06 (s, 1H), 5.44 (sept, J = 6.7 Hz, 1H), 2.95 (sept, J = 7 Hz, 1H), 1.45 (d, J = 6.7 Hz, 6H), 1.29 (d, J = 7 Hz, 6H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CDCl_3 , 25 °C): δ = 182.1, 152.1, 142.1, 117.2, 48.6, 28.0, 23.4, 22.4 ppm. MS (EI-methylene chloride): m/z 180.2 (14%), 165.2 (17), 123.2 (100).

4-*t*-butyl-1-isopropylimidazole-2-carbaldehyde: The oil was crystallized from a concentrated solution in diethyl ether, cooled to –15 °C overnight, affording clear colorless crystals (1.29 g, 88%). ^1H NMR (500 MHz, CDCl_3 , 25 °C): δ = 9.78 (s, 1H), 7.06 (s, 1H), 5.43 (sept, J = 6.7 Hz, 1H), 1.45 (d, J = 6.7 Hz, 6H), 1.32 (s, 9H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CDCl_3 , 25 °C): δ = 182.2, 155.2, 142.1, 116.5, 48.6, 32.1, 30.2, 23.4 ppm; MS (EI-methylene chloride): m/z 194.2 (16%), 179.2 (50), 137.2 (100).

General procedure for the synthesis of (4-alkyl-1-isopropylimidazol-2-yl) methanol compounds: The synthesis was adapted from a previously reported procedure for the preparation of (1-isopropylimidazol-2-yl) methanol.^[40] Lithium aluminum hydride (1.4 equiv.) was dissolved in dry THF at a concentration of 0.2 M in a round-bottomed flask under inert gas on a Schlenk line and cooled to 0 °C. The appropriate carbaldehyde (1 equiv.) was separately dissolved in dry THF to a concentration of 0.1 M and added dropwise to the stirred lithium aluminum hydride solution. This was allowed to stir for 5 min, then n mL DI H_2O , n mL 15% NaOH, and $3n$ mL DI H_2O were added via slow dropwise addition consecutively (where n equals the amount of LiAlH_4 used in grams; caution: significant gas evolution). The reaction was then allowed to stir for 2.5 h at room temperature. Anhydrous magnesium sulfate was added to the mixture, and then the solution was filtered. The solvent was removed via evaporation to give an oily residue, which could be triturated with hexane and further dried to give a white powder. In most cases, the material thus obtained was spectroscopically and analytically pure without additional purification; however, it could be further crystallized from ethyl acetate/hexane if needed.

1-isopropylimidazol-2-yl methanol: (1.34 g, 66%). ^1H NMR (500 MHz, CDCl_3 , 25 °C): δ = 6.91 (s, 1H), 6.87 (s, 1H), 4.67 (s, 2H), 4.66 (sept, J = 6.7 Hz, 1H), 1.45 (d, J = 6.7 Hz, 6H) ppm. Spectroscopic data was consistent with previous reports.^[40]

1,4-diisopropylimidazol-2-yl methanol: (1.105 g, 48%). ^1H NMR (500 MHz, CDCl_3): δ = 6.61 (s, 1H), 5.45 (bs, -OH, 1H), 4.65 (s, 2H), 4.56 (sept, J = 6.7 Hz, 1H), 2.80 (sept, J = 7 Hz, 1H), 1.43 (d, J = 6.7 Hz, 6H), 1.18 (d, J = 7 Hz, 6H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CDCl_3 , 25 °C): δ = 147.3, 146.1, 109.8, 55.7, 47.3, 27.7, 23.7, 22.4 ppm. MS (EI-methylene chloride): m/z 182.2 (41%), 167.2 (90), 125.1 (56), 107.1 (100); elemental analysis calcd (%) for $\text{C}_{10}\text{H}_{18}\text{N}_2\text{O}$: C, 65.90; H, 9.95; N, 15.37; found: C, 65.95; H, 10.19; N, 15.39.

4-*t*-butyl-1-isopropylimidazol-2-yl methanol: (1.02 g, 91%). ^1H NMR (500 MHz, CDCl_3 , 25 °C): δ = 6.59 (s, 1H), 5.29 (bs, -OH, 1H), 4.68 (s, 2H), 4.55 (sept, J = 6.6 Hz, 1H), 1.42 (d, J = 6.6 Hz, 6H), 1.19 (s, 9H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CDCl_3 , 25 °C): δ = 150.4, 145.5, 109.2, 56.4, 47.2, 31.6, 30.1, 23.7 ppm; MS (EI-methylene chloride): m/z 196.2 (15%), 167.2 (38), 121.1 (100); elemental analysis calcd (%) for $\text{C}_{11}\text{H}_{20}\text{N}_2\text{O}$: C, 67.31; H, 10.27; N, 14.27; found: C, 67.82; H, 10.57; N, 14.40.

General procedure for the synthesis of 2-chloromethyl-4-alkyl-1-isopropylimidazole hydrochloride compounds: The synthesis was adapted from a previously reported procedure for the preparation of 2-(chloromethyl)-1-isopropyl-1H-imidazole hydrochloride.^[40] The desired (4-alkyl-1-isopropylimidazol-2-yl) methanol (1 equiv.) was dissolved in DCM to a concentration of 0.05 M and thionyl chloride (1.9 equiv.) was added, and the reaction mixture was allowed to stir at room temperature overnight. The DCM was evaporated until roughly 10 mL of solution remained, and then 80 mL of ethyl acetate were added, resulting in the precipitation of an off-white to brown solid. The solid was collected on a frit, washed with diethyl ether, dried *in vacuo*, and used without further purification in subsequent steps.

2-chloromethyl-1-isopropylimidazole hydrochloride: (1.74 g, 93%) ^1H NMR (500 MHz, $\text{DMSO}-d_6$, 25 °C): δ = 8.03 (s, 1H), 7.77 (s, 1H), 5.22 (s, 2H), 4.80 (sept, J = 6.6 Hz, 1H), 1.47 (d, J = 6.6 Hz, 6H) ppm. Spectroscopic data was consistent with previous reports.^[62]

2-chloromethyl-1,4-diisopropylimidazole hydrochloride: In this case, the product was a viscous oil which did not produce a solid precipitate upon attempted recrystallization but appeared spectroscopically pure and was used without further purification for the next step (0.515 g, 79%). ^1H NMR (500 MHz, $\text{DMSO}-d_6$, 25 °C): δ = 7.79 (s, 1H), 5.20 (s, 2H), 4.78 (sept, J = 6.6 Hz, 1H), 2.98 (sept, J = 7 Hz, 1H), 1.46 (d, J = 6.6 Hz, 6H), 1.27 (d, J = 7 Hz, 6H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CDCl_3 , 25 °C): δ = 142.2, 139.8, 112.5, 50.3, 30.9, 25.6, 23.1, 21.6 ppm.

2-chloromethyl-4-*t*-butyl-1-isopropylimidazole hydrochloride: (0.186 g, 98%). ^1H NMR (500 MHz, $\text{DMSO}-d_6$, 25 °C): δ = 7.77 (s, 1H), 5.19 (s, 2H), 4.75 (sept, J = 6.7 Hz, 1H), 1.47 (d, J = 6.7 Hz, 6H) 1.32 (s, 9H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, $\text{DMSO}-d_6$, 25 °C): δ = 140.1, 114.4, 49.9, 31.6, 30.8, 29.0, 22.2 ppm; elemental analysis calcd (%) for $\text{C}_{11}\text{H}_{20}\text{Cl}_2\text{N}_2$: C, 52.60; H, 8.03; N, 11.15; found: C, 52.72; H, 8.19; N, 10.92.

General procedure for the synthesis of ligands $\text{R}_{\text{IP}}\text{L}$: (Method A) The generalized procedure was adapted from previously reported related systems.^[38,48] In a Schlenk flask, 1 equiv. of tacn was dissolved in dry MeCN to a concentration of about 0.02 M and 4 equiv. of the desired 2-chloromethyl-4-alkyl-1-isopropylimidazole hydrochloride salt were added along with a stir bar. The flask was capped with a septum and purged with inert gas. After purging, 8 equiv. of DIPEA were added, and the cloudy mixture became transparent. This was allowed to react overnight, then the MeCN was removed by rotary evaporation. The crude oil was dissolved in DCM, and the DCM solution was washed with 10% aqueous NH_3 three times. The DCM layer was dried with anhydrous sodium sulfate, filtered, and evaporated. (**Method B**) An alternative procedure could be used for the synthesis of $\text{R}_{\text{IP}}\text{L}$ via reductive amination of tacn using the 1-isopropyl-4-alkyl-imidazole-2-carbaldehydes produced in earlier steps.^[37] Tacn was dissolved in dichloroethane (DCE) at a 0.1 M concentration, 4.5 equiv. of sodium triacetoxyborohydride (STAB) were added, and the flask was placed under an inert atmosphere. The substituted imidazole-2-carbaldehyde (3.5 equiv.) in a small amount of DCE was then added dropwise. After the addition, acetic acid (1 equiv.) was added and the reaction mixture was allowed to stir for three days. The reaction was quenched by adding 15% NaOH (approximately half the reaction volume) to the reaction mixture, and the product was extracted using DCM. The DCM extracts were combined and washed with brine, then dried using anhydrous sodium sulfate. The sodium sulfate was filtered off and the organic layer was evaporated revealing the crude oil.

1,4,7-tris((1-isopropylimidazol-2-yl) methyl)-1,4,7-triazocyclononane ($\text{H}_{\text{IP}}\text{L}$): Prepared using *method A* (0.452 g, 71 %) and purified

on an alumina column using 2.5 % methanol in DCM as the eluent. ^1H NMR (500 MHz, CDCl_3 , 25 °C): δ = 6.94 (s, 3H), 6.92 (s, 3H), 4.71 (sept, J = 6.5 Hz, 3H), 3.62 (s, 6H), 2.7 (s, 12H), 1.39 (d, J = 6.5 Hz, 18H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CDCl_3 , 25 °C): δ = 144.5, 127.6, 55.0, 54.9, 46.9, 23.8 ppm; HRMS (ESI-positive) calcd for $[\text{NaC}_{27}\text{H}_{45}\text{N}_9]^+$ ($[\text{L} + \text{Na}]^+$): m/z 518.3690; found: m/z 518.3733; calcd for $[\text{C}_{27}\text{H}_{46}\text{N}_9]^+$ ($[\text{L} + \text{H}]^+$): m/z 496.3871; found: m/z 496.3929.

1,4,7-tris((1,4-diisopropylimidazol-2-yl) methyl)-1,4,7-triazocyclononane ($^{\text{IPr,IPr}}\text{L}$): Prepared using *method A* (0.163 g, 48 %). Following the final DCM evaporation, the crude oil was dissolved in 75 % MeOH:25 % H_2O mixture. The crude product was then purified by passage through a C-18 on silica reverse phase column using a gradient elution from 75:25 MeOH: H_2O to pure MeOH. ^1H NMR (500 MHz, CDCl_3 , 25 °C): δ = 6.59 (s, 3H), 4.68 (sept, J = 6.7 Hz, 3H), 3.59 (s, 6H), 2.81 (sept, J = 6.9 Hz, 3H) 2.66 (s, 12H), 1.36 (d, J = 6.7 Hz, 18H), 1.21 (d, J = 6.9 Hz, 18H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CDCl_3 , 25 °C): δ = 147.7, 143.6, 109.5, 55.4, 55.1, 46.7, 27.9, 23.7, 23.5 ppm; HRMS (ESI-positive) calcd for $[\text{C}_{36}\text{H}_{64}\text{N}_9]^+$ ($[\text{L} + \text{H}]^+$): m/z 622.5279; found: m/z 622.5180.

1,4,7-tris((1-isopropyl-4-*t*-butyl-imidazol-2-yl) methyl)-1,4,7-triazocyclononane ($^{\text{tBu,IPr}}\text{L}$): *Method A*: Following the final DCM evaporation, the crude oil was dissolved in a 50 % MeOH:50 % H_2O mixture. The crude product was then purified by passage through a C-18 on silica reverse phase column using a gradient elution from 1:1 MeOH: H_2O to pure MeOH, and the product was isolated as a clear viscous oil (0.3624 g, 37 %). *Method B*: The crude oil was purified by column chromatography on basic alumina with dichloromethane as the eluent (0.580 g, 30 %). ^1H NMR (500 MHz, CDCl_3 , 25 °C): δ = 6.58 (s, 3H), 4.68 (sept, J = 6.7 Hz, 3H), 3.60 (s, 6H), 2.62 (s, 12H), 1.36 (d, J = 6.7 Hz, 18H), 1.23 (s, 27H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CDCl_3 , 25 °C): δ = 150.8, 143.7, 108.9, 55.5, 55.2, 31.7, 30.3, 23.9 ppm; HRMS (ESI-positive) calcd for $[\text{C}_{39}\text{H}_{70}\text{N}_9]^+$ ($[\text{L} + \text{H}]^+$): m/z 664.5749; found: m/z 664.5747.

Generalized procedure for the synthesis of $[\text{R}^{\text{IPr}}\text{LMn}](\text{ClO}_4)_2$: To a flask with a stir bar, 1 equiv. of $\text{M}(\text{ClO}_4)_2 \cdot 6\text{H}_2\text{O}$ was added and dissolved in reagent grade ethanol at a concentration of about 0.01 M. Separately, 1 equiv. of $\text{R}^{\text{IPr}}\text{L}$ was dissolved in minimal ethanol. This was added to the reaction flask, resulting in an immediate slight darkening of the reaction mixture, followed shortly by the formation of a colorless precipitate. The reaction mixture was left to stir for one hour, then the ethanol was evaporated. The crude product was dissolved in methanol or acetonitrile, and the pure product precipitated by addition of diethyl ether to the solution. The powder was collected via filtration and was recrystallized from acetonitrile or methanol by diffusion of diethyl ether, resulting in colorless to slightly pale-yellow crystals.

Caution: Organometallic perchlorate compounds have the potential to be explosive. We have not encountered any such issues in the course of our study, but work with these compounds should be carried out only on a small scale and with appropriate precautions.

1,4,7-tris((1-isopropylimidazol-2-yl) methyl)-1,4,7-triazocyclononane]manganese(II) perchlorate ($^{\text{H,IPr}}\text{LMn}](\text{ClO}_4)_2$): Recrystallized from acetonitrile/ether (0.161 g, 53 %). μ_{eff} (Evans method, acetonitrile- d_3 , 25 °C): 5.7 μ_B ; HRMS (ESI-positive) calcd for $[\text{C}_{27}\text{H}_{45}\text{ClMnN}_9\text{O}_4]^+$ ($[\text{M}-\text{ClO}_4]^+$): m/z 649.2658; found: m/z 649.2715; elemental analysis calcd for $\text{C}_{27}\text{H}_{45}\text{Cl}_2\text{MnN}_9\text{O}_8$: C, 43.27; H, 6.05; N, 16.82; found: C, 43.26; H, 6.067; N, 16.88.

1,4,7-tris((1-isopropylimidazol-2-yl) methyl)-1,4,7-triazocyclononane]zinc(II) perchlorate ($^{\text{H,IPr}}\text{LZn}](\text{ClO}_4)_2$): Recrystallized from acetonitrile/ether (0.099 g, 66 %). ^1H NMR (300 MHz, CD_3CN , 25 °C): δ = 7.25 (s, 3H), 6.58 (s, 3H), 4.37 (sept, J = 7 Hz, 3H), 2.98 (m, 6H), 2.81 (m, 6H), 1.41 (d, J = 7 Hz, 18H); $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CD_3CN , 25 °C): δ = 145.5, 125.2, 119.3, 53.0, 51.2, 48.9,

22.6; HRMS (ESI-positive) calcd for $[\text{C}_{27}\text{H}_{45}\text{ClZnN}_9\text{O}_4]^+$ ($[\text{M}-\text{ClO}_4]^+$): m/z 658.2569; found: m/z 658.2989; elemental analysis calcd for $\text{C}_{27}\text{H}_{45}\text{Cl}_2\text{ZnN}_9\text{O}_8$: C, 42.67; H, 5.97; N, 16.59; found: C, 42.84; H, 6.051; N, 16.54.

1,4,7-tris((1,4-diisopropylimidazol-2-yl) methyl)-1,4,7-triazocyclononane]manganese(II) perchlorate ($^{\text{IPr,IPr}}\text{LMn}](\text{ClO}_4)_2$): Recrystallized from acetonitrile/ether as a bis(acetonitrile) solvate; the cocrystallized acetonitrile is removed by crushing up the crystals and drying *in vacuo* overnight (0.055 g, 49 %). μ_{eff} (Evans method, acetonitrile- d_3 , 25 °C): 6.2 μ_B ; HRMS (ESI-positive) calcd for $[\text{C}_{36}\text{H}_{63}\text{ClMnN}_9\text{O}_4]^+$ ($[\text{M}-\text{ClO}_4]^+$): m/z 775.4067; found: m/z 775.4087; calcd for $[\text{C}_{36}\text{H}_{63}\text{MnN}_9]^{2+}$ ($[\text{M}-2\text{ClO}_4]^{2+}$): m/z 338.2288; found: m/z 338.2313; elemental analysis calcd for $\text{C}_{36}\text{H}_{63}\text{Cl}_2\text{MnN}_9\text{O}_8$: C, 49.37; H, 7.25; N, 14.39; found: C, 49.78; H, 7.37; N, 13.84.

1,4,7-tris((1,4-diisopropylimidazol-2-yl) methyl)-1,4,7-triazocyclononane]zinc(II) perchlorate ($^{\text{IPr,IPr}}\text{LZn}](\text{ClO}_4)_2$): Recrystallized from methanol/ether as a bis(methanol) solvate (see crystallographic details); methanol is removed by crushing up the crystals and drying *in vacuo* overnight (0.036 g, 32 %). ^1H NMR (500 MHz, CD_3CN , 25 °C): δ = 7.09 (s, 3H), 4.37 (sept, J = 6.6 Hz, 3H), 4.22 (d, 3H), 3.90 (d, 3H), 3.15 (m, 3H), 2.78 (m, 9H), 2.02 (sept, J = 7 Hz, 3H), 1.42 (d, J = 6.6 Hz, 9H), 1.34 (d, J = 6.6 Hz, 9H), 1.06 (d, J = 7 Hz, 9H), 0.62 (d, J = 7 Hz, 9H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (126 MHz, CD_3CN , 25 °C): δ = 147.7, 144.3, 114.3, 53.6, 52.4, 50.8, 48.9, 27.5, 24.5, 22.7, 22.5, 21.8 ppm; HRMS (ESI-positive) calcd for $[\text{C}_{36}\text{H}_{63}\text{ClZnN}_9\text{O}_4]^+$ ($[\text{M}-\text{ClO}_4]^+$): m/z 784.3978; found: m/z 784.4153; calcd for $[\text{C}_{36}\text{H}_{63}\text{ZnN}_9]^{2+}$ ($[\text{M}-2\text{ClO}_4]^{2+}$): m/z 342.7243; found: m/z 342.7330; elemental analysis calcd for $\text{C}_{36}\text{H}_{63}\text{Cl}_2\text{ZnN}_9\text{O}_8$: C, 48.79; H, 7.17; N, 14.22; found: C, 49.10; H, 7.30; N, 13.85.

1,4,7-tris((1-isopropyl-4-*tert*-butylimidazol-2-yl) methyl)-1,4,7-triazocyclononane]manganese(II) perchlorate ($^{\text{tBu,IPr}}\text{LMn}](\text{ClO}_4)_2$): Recrystallized from acetonitrile/ether (0.033 g, 12 %). μ_{eff} (Evans method, acetonitrile- d_3 , 25 °C): 5.9 μ_B ; HRMS (ESI-positive) calcd for $[\text{C}_{39}\text{H}_{69}\text{ClMnN}_9\text{O}_4]^+$ ($[\text{M}-\text{ClO}_4]^+$): m/z 817.4536; found: m/z 817.4835; calcd for $[\text{C}_{39}\text{H}_{69}\text{MnN}_9]^{2+}$ ($[\text{M}-2\text{ClO}_4]^{2+}$): m/z 359.2523; found: m/z 359.2583; elemental analysis calcd for $\text{C}_{39}\text{H}_{69}\text{Cl}_2\text{MnN}_9\text{O}_8$: C, 51.03; H, 7.58, N, 13.73; found: C, 51.04; H, 7.708; N, 13.72.

1,4,7-tris((1-isopropyl-4-*tert*-butylimidazol-2-yl) methyl)-1,4,7-triazocyclononane]zinc(II) perchlorate ($^{\text{tBu,IPr}}\text{LZn}](\text{ClO}_4)_2$): Recrystallized from acetonitrile/ether (0.123 g, 50 %). ^1H NMR (300 MHz, CD_3CN , 55 °C): δ = 7.18 (s, 3H), 4.48 (sept, J = 6 Hz, 3H), 4.19 (br, 6H), 3.09 (br, 6H), 2.69 (br, 6H), 1.44 (d, J = 6 Hz, 18H), 1.00 (s, 27H) ppm; $^{13}\text{C}\{^1\text{H}\}$ NMR (150 MHz, CD_3CN , 25 °C): δ = 151.7, 145.3, 116.5, 57.2, 55.6 (br), 53.7 (br), 49.3, 32.4, 31.6, 23.1 (br) ppm; HRMS (ESI-positive) calcd for $[\text{C}_{39}\text{H}_{69}\text{ClZnN}_9\text{O}_4]^+$ ($[\text{M}-\text{ClO}_4]^+$): m/z 826.4447; found: m/z 826.5201; calcd for $[\text{C}_{39}\text{H}_{69}\text{ZnN}_9]^{2+}$ ($[\text{M}-2\text{ClO}_4]^{2+}$): m/z 363.7478; found: m/z 363.7606; elemental analysis calcd for $\text{C}_{39}\text{H}_{69}\text{Cl}_2\text{N}_9\text{O}_8\text{Zn}$: C, 50.46; H, 7.49; N, 13.59; found: C, 51.01; H, 7.79; N, 13.14.

Author Contributions

RBG carried out the synthesis of ligands and complexes with assistance from BNM, SLL, KAC, WES, DEM, GM, and SEC. RBG carried out spectroscopic and electrochemical analyses. BD performed X-ray single crystal analysis. SEC, KAC, and RBG carried out binding constant determinations. RBG carried out DFT computations. SEC and WES carried out cavity size calculations. SEC and RBG drafted the manuscript and all authors contributed to editing the manuscript. SEC directed the project and obtained funding.

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Conflict of Interests

There are no conflicts to declare.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

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- [1] K. J. Waldron, N. J. Robinson, *Nat. Rev. Microbiol.* **2009**, *7*, 25–35.
- [2] J. Park, M. B. Cleary, D. Li, J. A. Mattocks, J. Xu, H. Wang, S. Mukhopadhyay, E. M. Gale, J. A. Cotruvo, *Proc. Natl. Acad. Sci.* **2022**, *119*, e2212723119.
- [3] S. Bakthavatsalam, A. Sarkar, A. Rakshit, S. Jain, A. Kumar, A. Datta, *Chem. Commun.* **2015**, *51*, 2605–2608.
- [4] S. Das, K. Khatua, A. Rakshit, A. Carmona, A. Sarkar, S. Bakthavatsalam, R. Ortega, A. Datta, *Dalton Trans.* **2019**, *48*, 7047–7061.
- [5] B. Drahoš, I. Lukeš, É. Tóth, *Eur. J. Inorg. Chem.* **2012**, *2012*, 1975–1986.
- [6] E. M. Gale, I. P. Atanasova, F. Blasi, I. Ay, P. Caravan, *J. Am. Chem. Soc.* **2015**, *137*, 15548–15557.
- [7] P. Cieslik, P. Comba, B. Dittmar, D. Ndiaye, É. Tóth, G. Velmurugan, H. Wadepohl, *Angew. Chem. Int. Ed.* **2022**, *61*, e202115580.
- [8] D. Ndiaye, P. Cieslik, H. Wadepohl, A. Pallier, S. Mème, P. Comba, É. Tóth, *J. Am. Chem. Soc.* **2022**, *144*, 22212–22220.
- [9] E. D. Weinberg, *J. Am. Med. Assoc.* **1975**, *231*, 39.
- [10] M. I. Hood, E. P. Skaar, *Nat. Rev. Microbiol.* **2012**, *10*, 525–537.
- [11] C. C. Murdoch, E. P. Skaar, *Nat. Rev. Microbiol.* **2022**, *10.1038/s41579-022-00745-6*.
- [12] P. A. Hessian, J. Edgeworth, N. Hogg, *J. Leukocyte Biol.* **1993**, *53*, 197–204.
- [13] B. A. Gilston, E. P. Skaar, W. J. Chazin, *Sci. China: Life Sci.* **2016**, *59*, 792–801.
- [14] E. M. Zygiel, E. M. Nolan, *Annu. Rev. Biochem.* **2018**, *87*, 621–643.
- [15] R. C. Hadley, D. M. Gagnon, M. B. Brophy, Y. Gu, T. G. Nakashige, R. D. Britt, E. M. Nolan, *J. Am. Chem. Soc.* **2018**, *140*, 110–113.
- [16] J. R. Stephan, E. M. Nolan, *Chem. Sci.* **2016**, *7*, 1962–1975.
- [17] D. M. Gagnon, M. B. Brophy, S. E. J. Bowman, T. A. Stich, C. L. Drennan, R. D. Britt, E. M. Nolan, *J. Am. Chem. Soc.* **2015**, *137*, 3004–3016.
- [18] R. C. Hadley, D. M. Gagnon, A. Ozarowski, R. D. Britt, E. M. Nolan, *Inorg. Chem.* **2019**, *58*, 13578–13590.
- [19] I. P. Korndörfer, F. Brueckner, A. Skerra, *J. Mol. Biol.* **2007**, *370*, 887–898.
- [20] B. D. Corbin, E. H. Seeley, A. Raab, J. Feldmann, M. R. Miller, V. J. Torres, K. L. Anderson, B. M. Dattilo, P. M. Dunman, R. Gerads, R. M. Caprioli, W. Nacken, W. J. Chazin, E. P. Skaar, *Science* **2008**, *319*, 962–965.
- [21] M. B. Brophy, E. M. Nolan, *ACS Chem. Biol.* **2015**, *10*, 641–651.
- [22] E. M. Zygiel, E. M. Nolan, *Acc. Chem. Res.* **2019**, *52*, 2301–2308.
- [23] T. G. Nakashige, E. M. Zygiel, C. L. Drennan, E. M. Nolan, *J. Am. Chem. Soc.* **2017**, *139*, 8828–8836.
- [24] T. E. Kehl-Fie, S. Chitayat, M. I. Hood, S. Damo, N. Restrepo, C. Garcia, K. A. Munro, W. J. Chazin, E. P. Skaar, *Cell Host Microbe* **2011**, *10*, 158–164.
- [25] S. Damo, W. J. Chazin, E. P. Skaar, T. E. Kehl-Fie, *Virulence* **2012**, *3*, 325–328.
- [26] S. M. Damo, T. E. Kehl-Fie, N. Sugitani, M. E. Holt, S. Rath, W. J. Murphy, Y. Zhang, C. Betz, L. Hench, G. Fritz, E. P. Skaar, W. J. Chazin, *Proc. Nat. Acad. Sci.* **2013**, *110*, 3841–3846.
- [27] T. G. Nakashige, J. R. Stephan, L. S. Cunden, M. B. Brophy, A. J. Wommack, B. C. Keegan, J. M. Shearer, E. M. Nolan, *J. Am. Chem. Soc.* **2016**, *138*, 12243–12251.
- [28] H. Irving, R. J. P. Williams, *Nature* **1948**, *162*, 746–747.
- [29] H. Irving, R. J. P. Williams, *J. Chem. Soc.* **1953**, 3192.
- [30] J. J. Griesse, K. Roos, N. Cox, H. S. Shafaat, R. M. M. Branca, J. Lehtiö, A. Gräslund, W. Lubitz, P. E. M. Siegbahn, M. Högbom, *Proc. Nat. Acad. Sci.* **2013**, *110*, 17189–17194.
- [31] E. C. Kisgeropoulos, J. J. Griesse, Z. R. Smith, R. M. M. Branca, C. R. Schneider, M. Högbom, H. S. Shafaat, *J. Am. Chem. Soc.* **2020**, *142*, 5338–5354.
- [32] K. Gräve, J. J. Griesse, G. Berggren, M. D. Bennett, M. Högbom, *J. Biol. Inorg. Chem.* **2020**, *25*, 571–582.
- [33] R. D. Shannon, *Acta Crystallogr. Sect. A* **1976**, *32*, 751–767.
- [34] A. Hu, E. Aluicio-Sarduy, V. Brown, S. N. MacMillan, K. V. Becker, T. E. Barnhart, V. Radchenko, C. F. Ramogida, J. W. Engle, J. J. Wilson, *J. Am. Chem. Soc.* **2021**, *143*, 10429–10440.
- [35] P. B. Tsitovich, A. M. Kossawattarachchi, M. R. Crawley, T. Y. Tittiris, T. R. Cook, J. R. Morrow, *Chem. Eur. J.* **2017**, *23*, 15327–15331.
- [36] C. Götzmänn, F. Braun, M. D. Bartholomä, *RSC Adv.* **2015**, *6*, 119–131.
- [37] P. B. Tsitovich, F. Gendron, A. Y. Nazarenko, B. N. Livesay, A. P. Lopez, M. P. Shores, J. Autschbach, J. R. Morrow, *Inorg. Chem.* **2018**, *57*, 8364–8374.
- [38] M. D. Vaira, F. Mani, P. Stoppioni, *J. Chem. Soc. Chem. Commun.* **1989**, 126–127.
- [39] T. N. Sorrell, W. E. Allen, *J. Org. Chem.* **1994**, *59*, 1589–1590.
- [40] M. Seto, N. Miyamoto, K. Aikawa, Y. Aramaki, N. Kanzaki, Y. Iizawa, M. Baba, M. Shirashi, *Bioorg. Med. Chem.* **2005**, *13*, 363–386.
- [41] Deposition Number(s) 2218031 (for $[^{\text{H}}\text{Pr}_1\text{LMn}]$), 2218032 (for $[^{\text{Pr}}\text{Pr}_1\text{LMn}]$), 2218028 (for $[^{\text{Bu}}\text{Pr}_1\text{LMn}]$), 2218029 (for $[^{\text{H}}\text{Pr}_1\text{LZn}]$), 2218030 (for $[^{\text{Pr}}\text{Pr}_1\text{LZn}]$), 2218033 (for $[^{\text{Bu}}\text{Pr}_1\text{LZn}]$) contain(s) the supplementary crystallographic data for this paper. These data are provided free of charge by the joint Cambridge Crystallographic Data Centre and Fachinformationszentrum Karlsruhe Access Structures service.
- [42] C. S. Slone, D. A. Weinberger, C. A. Mirkin, in *Progress in Inorganic Chemistry*, John Wiley & Sons, Ltd, **1999**, pp. 233–350.
- [43] S. Romain, C. Duboc, F. Neese, E. Rivière, L. R. Hanton, A. G. Blackman, C. Philouze, J.-C. Leprêtre, A. Deronzier, M.-N. Collomb, *Chem. Eur. J.* **2009**, *15*, 980–988.
- [44] C. F. Elliott, K. E. Fraser, S. A. Odom, C. Risko, *J. Phys. Chem. A* **2021**, *125*, 272–278.
- [45] P. Hosseinzadeh, Y. Lu, *Biochim. Biophys. Acta Bioenerg.* **2016**, *1857*, 557–581.
- [46] Rita. Yang, L. J. Zompa, *Inorg. Chem.* **1976**, *15*, 1499–1502.
- [47] M. Roger, L. M. P. Lima, M. Frindel, C. Platas-Iglesias, J.-F. Gestin, R. Delgado, V. Patinec, R. Tripier, *Inorg. Chem.* **2013**, *52*, 5246–5259.
- [48] L. Tei, A. Bencini, A. J. Blake, V. Lippolis, A. Perra, B. Valtancoli, C. Wilson, M. Schröder, *Dalton Trans.* **2004**, 1934–1944.
- [49] D. Ndiaye, M. Sy, A. Pallier, S. Mème, I. de Silva, S. Lacerda, A. M. Nonat, L. J. Charbonnière, É. Tóth, *Angew. Chem. Int. Ed.* **2020**, *59*, 11958–11963; *Angew. Chem.* **2020**, *132*, 12056–12061.
- [50] E. Molnár, B. Váradi, Z. Garda, R. Botár, F. K. Kálmán, É. Tóth, C. Platas-Iglesias, I. Tóth, E. Brucher, G. Tircsó, *Inorg. Chim. Acta* **2018**, *472*, 254–263.
- [51] J. Liang, J. W. Canary, *Angew. Chem. Int. Ed.* **2010**, *49*, 7710–7713; *Angew. Chem.* **2010**, *122*, 7876–7879.
- [52] P. Comba, W. Schiek, *Coord. Chem. Rev.* **2003**, *238–239*, 21–29.
- [53] P. Comba, M. Kerscher, M. Merz, V. Müller, H. Pritzkow, R. Remenyi, W. Schiek, Y. Xiong, *Chem. Eur. J.* **2002**, *8*, 5750–5760.
- [54] M. J. Frisch, G. W. Trucks, H. B. Schlegel, G. E. Scuseria, M. A. Robb, J. R. Cheeseman, G. Scalmani, V. Barone, G. A. Petersson, H. Nakatsuji, X. Li, M. Caricato, A. V. Marenich, J. Bloino, B. G. Janesko, R. Gomperts, B. Mennucci, H. P. Hratchian, J. V. Ortiz, A. F. Izmaylov, J. L. Sonnenberg, Williams, F. Ding, F. Lipparini, F. Egidi, J. Goings, B. Peng, A. Petrone, T. Henderson, D. Ranasinghe, V. G. Zakrzewski, J. Gao, N. Rega, G. Zheng, W. Liang, M. Hada, M. Ehara, K. Toyota, R. Fukuda, J. Hasegawa, M. Ishida, T. Nakajima, Y. Honda, O. Kitao, H. Nakai, T. Vreven, K. Throssell, J. A. Montgomery Jr., J. E. Peralta, F. Ogliaro, M. J. Bearpark, J. J. Heyd, E. N. Brothers, K. N. Kudin, V. N. Staroverov, T. A. Keith, R. Kobayashi, J. Normand, K. Raghavachari, A. P. Rendell, J. C. Burant, S. S. Iyengar, J. Tomasi, M. Cossi, J. M. Millam, M. Klene, C. Adamo, R. Cammi, J. W.

- Ochterski, R. L. Martin, K. Morokuma, O. Farkas, J. B. Foresman, D. J. Fox, Gaussian, Inc., Wallingfor CT, 2016.
- [55] S. Grimme, J. Antony, S. Ehrlich, *J. Chem. Phys.* **2010**, *132*, 154104.
- [56] T. N. Sorrell, W. E. Allen, G. Parkin, C. Kimblin, in *Inorganic Syntheses* (Ed.: Marcetta Y. Darensbourg), John Wiley & Sons, Inc., Hoboken, NJ, USA, **2007**, pp. 66–70.
- [57] M. Gaudry, A. Marquet, *Org. Synth.* **1976**, *55*, 24.
- [58] M. Moir, R. Boyd, H. Gunosewoyo, A. P. Montgomery, M. Connor, M. Kassiou, *Tetrahedron Lett.* **2019**, *60*, 151019.
- [59] S. E. Howson, L. E. N. Allan, N. P. Chmel, G. J. Clarkson, R. J. Deeth, A. D. Faulkner, D. H. Simpson, P. Scott, *Dalton Trans.* **2011**, *40*, 10416.
- [60] N. Ferri, S. Cazzaniga, L. Mazzarella, G. Curigliano, G. Lucchini, D. Zerla, R. Gandolfi, G. Facchetti, M. Pellizzoni, I. Rimoldi, *Bioorg. Med. Chem.* **2013**, *21*, 2379–2386.
- [61] N. Samaan, Q. Zhong, J. Fernandez, G. Chen, A. M. Hussain, S. Zheng, G. Wang, Q.-H. Chen, *Eur. J. Med. Chem.* **2014**, *75*, 123–131.
- [62] A. Tasaka, K. Teranishi, Y. Matsushita, N. Tamura, R. Hayashi, K. Okonogi, K. Itoh, *Chem. Pharm. Bull.* **1994**, *42*, 85–94.

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